

Virtual LANs and Traffic Segmentation

A virtual LAN, or VLAN, groups switch ports into separate broadcast domains even when they share the same physical switch. IEEE 802.1Q inserts a tag into the Ethernet frame on trunk links. Access ports belong to a single VLAN and send untagged frames to workstations.

Segmentation by VLAN limits broadcast traffic and supports security policy. Finance workstations, laboratory instruments, IP telephones, and guest laptops can share one switch stack yet remain isolated. Inter-VLAN communication requires a router or a layer-3 switch. Access control lists on that device decide which subnets may talk to each other.

Poor VLAN design recreates a flat network. Placing every port in VLAN 1, using the native VLAN for user data, or allowing user ports to negotiate trunking opens the door to VLAN hopping. Best practice is a dedicated unused native VLAN, explicit trunk allow-lists, and unused ports parked in a black-hole VLAN.

Voice VLANs and data VLANs on the same access port are common. The telephone uses the tagged voice VLAN, while the PC behind the telephone uses the untagged data VLAN. Quality of service markings then protect latency-sensitive voice packets across the campus LAN.